

A PROJECT REPORT
ON
RADAR SYSTEM USING ARDUINO UNO BOARD

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THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF
THIRD YEAR COMPUTER ENGINEERING

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CERTIFICATE

This is to certify that the mini project titled

RADAR SYSTEM USING ARDUINO UNO BOARD

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DECLARATION

I **Siddhi Sandeep Bedge** hereby declare that the mini project report **Radar System using Arduino Board**. I, submitted for partial fulfillment of the requirements for the award of degree of Third Year in Computer Engineering of the NMVP and PCET's Nutan Maharashtra Institute of Engineering and Technology, Talegaon is a bonafide work done by me under supervision of **Dr. Rohini Hanchate**.

This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources.

I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the Institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

Project presents a low-cost radar system developed using an Arduino Uno microcontroller to detect nearby objects and visualize their positions in real time through the Processing IDE. The system integrates an HC-SR04 ultrasonic sensor mounted on a servo motor that rotates across a 180° field of view, enabling continuous environmental scanning. Distance measurements are obtained by calculating the time of flight of ultrasonic waves reflected from surrounding objects. The measured data, along with the servo angle, are transmitted to the Processing interface via serial communication, where detected objects are represented as red blips, and the scanning direction is shown as a green rotating line, creating a dynamic radar-like display. The system effectively detects objects within a 15 cm range and provides visibility up to 40 cm, demonstrating the core principles of short-range radar operation. This project highlights the integration of embedded electronics, sensor interfacing, automation, and real-time data visualization. It serves as a valuable educational and prototyping platform for understanding object detection, embedded system design, and interactive visualization techniques.

Keywords: Arduino Uno, Ultrasonic Sensor, Radar System, Real-Time Visualization, Embedded Systems, Object Detection, Processing IDE

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Chapter 1

INTRODUCTION

Radar systems are widely used for detecting and locating objects by measuring the time it takes for transmitted signals to reflect back from a surface. Conventional radar systems use radio waves, but in this project, we replicate the same concept using ultrasonic sound waves to design a low-cost radar model suitable for educational and experimental purposes.

The Radar System using Arduino Uno combines the capabilities of an HC-SR04 ultrasonic distance sensor, a servo motor, and an Arduino Uno microcontroller to create a scanning mechanism that detects objects within a specific range. The ultrasonic sensor emits sound pulses and measures the time it takes for the echo to return, calculating the distance of the detected object. The servo motor rotates the sensor from 15° to 165°, allowing the radar to scan the surrounding area in a semi-circular motion.

The measured distance and angle data are then transmitted to a computer via serial communication, where the Processing IDE software visually represents the information as a radar screen. This visualization resembles a real radar display, showing the position of detected objects as red dots and the scanning beam as a moving green line.

The project serves as a practical demonstration of several core engineering concepts — sensor interfacing, embedded programming, data processing, and real-time visualization. It highlights how microcontrollers can be used for automation, environment mapping, and object detection.

In real-world applications, such systems can be utilized for robotic navigation, security and surveillance systems, collision avoidance, and proximity detection. The simplicity of the components and the open-source nature of the software make this project both cost-effective and highly educational, offering students hands-on experience with electronics and programming.

1.1 BACKGROUND / OVERVIEW OF TOPIC

Radar, short for Radio Detection and Ranging, is a technology widely used in military, aviation, marine, and industrial applications to detect, locate, and track objects. Traditional radar systems operate by emitting radio waves that reflect off objects, with the reflected signals analyzed to determine the distance, position, and movement of targets. These systems are often expensive, complex, and require specialized hardware.

In recent years, the principles of radar technology have been adapted for educational and prototype applications using low-cost microcontrollers and sensors. This project focuses on developing a miniature radar system using an Arduino Uno and an HC-SR04 ultrasonic sensor, which employs ultrasonic sound waves instead of radio waves to measure distances.

The ultrasonic sensor is mounted on a servo motor, which sweeps the sensor across an angular range of 15° to 165° , allowing it to scan a semicircular area. As the sensor rotates, it sends distance measurements to the Arduino, which then communicates the data to a Processing IDE application on a computer. The Processing software visualizes the data in real-time radar format, displaying detected objects as red blips and the scanning direction as a moving green line.

This approach provides a cost-effective and educational demonstration of radar principles, enabling users to understand object detection, distance measurement, and real-time visualization without the complexity of conventional radar systems. The project also introduces learners to Arduino programming, sensor interfacing, serial communication, and graphical data representation, making it a comprehensive learning tool in electronics and embedded systems.

1.2 PROBLEM STATEMENT

Traditional radar systems are expensive, large, and technically complex, making them unsuitable for small-scale or educational use. The main challenge of this project is to design and develop a low-cost radar model that can detect and visualize nearby objects effectively using simple components and open-source tools.

The key issues addressed in this project are:

1. **High Cost of Conventional Systems:** Real radar systems use radio frequency modules and advanced signal processors, which are costly and difficult for students or hobbyists to implement.
2. **Complex Hardware and Circuitry:** Commercial radar systems require specialized antennas, transmitters, and receivers that involve complex design and calibration.
3. **Limited Accessibility for Learning:** Due to their complexity, radar systems are not easily available for basic learning, experimentation, or small-scale applications.
4. **Need for a Simple Educational Model:** There is a need for a compact, affordable system that demonstrates the working principles of radar using basic sensors and microcontrollers.
5. **Integration of Detection and Visualization:** The goal is not only to detect objects but also to visually represent their position in real time, helping users understand how scanning and distance mapping work together.
6. **Use of Readily Available Components:** The system should be developed using easily available components like Arduino Uno, an ultrasonic sensor, and a servo motor, reducing both cost and complexity.
7. **Open-Source Software Utilization:** By using Arduino IDE for programming and Processing IDE for data visualization, the entire setup becomes open, modifiable, and easy to replicate for future development or learning purposes.

1.3 OBJECTIVES

The main aim of this project is to design and develop a cost-effective radar system using Arduino that can detect nearby objects and display their position on a computer screen in real time. Each objective of the project is explained below

1. To design and implement a radar system using Arduino Uno:

The first objective focuses on building the core radar system using the Arduino Uno microcontroller. Arduino acts as the brain of the system — controlling the ultrasonic sensor, reading the distance data, and communicating with the Processing software. This step involves programming the Arduino to handle input-output operations and execute continuous scanning cycles efficiently.

2. To integrate an ultrasonic distance sensor for object detection:

The HC-SR04 ultrasonic sensor is used to measure the distance between the sensor and any object in front of it. It works by transmitting ultrasonic pulses and calculating the time taken for the echo to return. This sensor allows the system to detect obstacles with good accuracy at short ranges and acts as the main sensing component of the radar.

3. To rotate the sensor using a servo motor to scan a wide area:

The servo motor is responsible for rotating the ultrasonic sensor from 15° to 165° , enabling the radar to scan a semicircular area. This rotational motion helps in covering a wider detection field instead of focusing on a single point. It simulates the sweeping motion seen in real radar systems, making the detection process dynamic and more informative.

4. To visualize real-time distance and angle data using Processing IDE:

The Processing IDE is used to create a graphical user interface (GUI) that displays radar-like visuals. The Arduino sends the angle and distance values through serial communication, and the Processing program converts this data into a real-time radar display. This allows users to observe detected objects as red dots on a green radar screen, similar to professional radar systems.

5. To detect objects within a range of 40 cm accurately:

The radar is calibrated to accurately detect and display objects that are within 15 cm of the sensor. This range is ideal for demonstration purposes, providing quick and visible results on the screen. The system ensures reliable detection within this range, showcasing how short-distance radar systems can function effectively using simple components.

1.4 SCOPE OF THE STUDY

The Radar System using Arduino Uno has a wide scope in both educational and practical fields. It serves as a foundation for understanding how radar technology works and how it can be implemented using low-cost, open-source tools. The scope of this study includes the following aspects:

1. Educational Learning and Skill Development : This project provides students with hands-on experience in working with microcontrollers, sensors, and servo motors. It enhances understanding of Arduino programming, sensor interfacing, and serial communication, helping learners grasp fundamental concepts of embedded systems and automation.
2. Application in Robotics and Automation : The system can be used in robotic sensing for obstacle detection and navigation. Robots equipped with similar radar systems can identify obstacles in their path and take corrective actions, improving their ability to operate autonomously.
3. Proximity Detection and Obstacle Avoidance : The radar can be employed in small-scale applications such as parking sensors, door monitoring systems, or security setups, where proximity detection and short-range obstacle avoidance are required.
4. Real-Time Data Visualization : By integrating Arduino with the Processing IDE, the project demonstrates how sensor data can be converted into a graphical, real-time radar display. This offers a clear understanding of how raw data can be visualized for better analysis and interpretation.
5. Research and Future Development
The project serves as a base model for advanced systems that can use technologies like LiDAR,

infrared sensors, or machine learning for enhanced object detection. It can also be extended to include wireless communication or IoT-based monitoring for more advanced research projects.

6. Cost-Effective Prototype for Demonstration The system is designed using readily available, low-cost components, making it ideal for demonstration and laboratory experiments. It provides a simple yet effective way to simulate radar functionality without the complexity and expense of traditional radar systems.

Chapter 2

LITERATURE SURVEY

2.1 REVIEW OF PREVIOUS WORK

In recent years, there has been significant interest in developing low-cost radar and obstacle detection systems using microcontrollers and affordable sensors. Researchers and hobbyists have explored various approaches to replicate radar functionality for educational, prototype, and small-scale practical applications.

- **Ultrasonic Sensor-Based Radars:** Several projects have implemented radar systems using HC-SR04 ultrasonic sensors mounted on servo motors. These systems measure distance by calculating the time-of-flight of ultrasonic waves and visualize detected objects using software interfaces like Processing IDE or MATLAB. Such designs are widely used for obstacle detection, robotic navigation, and indoor mapping.
- **Infrared Sensor-Based Systems:** Some low-cost radar prototypes utilize infrared (IR) sensors for short-range detection. IR-based systems provide fast detection but are often limited by line-of-sight requirements and ambient light interference.
- **Applications:**
Existing Arduino-based radar projects have been applied in:
 - Security and surveillance systems: detecting intrusions within a defined area.
 - Robotic navigation: helping autonomous robots avoid obstacles.
 - Educational demonstrations: teaching radar principles in laboratories and workshops.
- **Software Integration:** Most prior work emphasizes real-time visualization, where detected objects are plotted on a radar-like interface, providing immediate feedback for analysis and demonstration.
- **Limitations Noted in Previous Work:** While these projects successfully demonstrate radar functionality, they are generally limited by short detection range, sensor accuracy, and environmental influences. Additionally, most prior implementations focus only on single-sensor scanning, lacking features like data logging, wireless transmission, or advanced object tracking.

Table 2.1: Literature Survey

Sr. No.	Author / Source	Method / Technology	Key Features / Findings	Limitations	Applications / Remarks
1	Arduino Projects (Processing.org, CircuitDigest)	HC-SR04 + Servo + Arduino Uno	Real-time radar visualization using Processing IDE.	Short range (~40 cm); noise sensitive.	Educational use, obstacle detection.
2	TutorialsPoint, Instructables	IR Sensor-Based Radar	Fast short-range detection with simple setup.	Affected by ambient light; requires line-of-sight.	Motion sensing, proximity detection.
3	Academic Prototypes	Microcontroller + Ultrasonic / IR Sensors	Low-cost radar simulation for security monitoring.	Limited range and moderate accuracy.	Security systems, indoor navigation.
4	MATLAB / Processing Studies	PC-Based Visualization	Graphical display of detected objects on a computer.	Requires wired link; minor processing delay.	Data analysis, demonstration purposes.
5	IoT-Based Projects (2022–2024)	Arduino + Wi-Fi / Bluetooth	Enables wireless transmission of radar data.	Higher power consumption and cost.	IoT applications, remote surveillance.

2.2 KEY THEORIES AND CONCEPTS

The Arduino-based radar system relies on several fundamental theories and concepts from electronics, physics, and embedded systems. Understanding these concepts is essential for designing, implementing, and analyzing the radar prototype.

1. Ultrasonic Distance Measurement

- Principle: Ultrasonic sensors work on the principle of echolocation, similar to how bats detect objects. The sensor emits ultrasonic sound waves (typically around 40 kHz) and measures the time interval for the echo to return after hitting an object.
- Distance Calculation: The distance is calculated using the formula:

$$\text{Distance} = \frac{\text{Time of Flight} \times \text{Speed of Sound}}{2}$$

The division by 2 accounts for the round trip of the sound wave (to the object and back).

- Application in the Project: The HC-SR04 sensor measures distances within a short range (up to 40 cm), providing real-time data to the Arduino.

2. Servo Motor Operation

- Principle: Servo motors are rotational actuators that move to a specified angle when given a control signal (PWM).
- PWM Control: By sending Pulse Width Modulation (PWM) signals from the Arduino, the servo can rotate precisely to any angle within its mechanical range (e.g., 0°–180°).
- Application in the Project: The servo rotates the ultrasonic sensor from 15° to 165°, enabling scanning across a semicircular area for object detection.

3. Serial Communication

- Principle: Serial communication allows data exchange between two devices, such as the Arduino and a computer. Data is transmitted bit by bit over a communication line.
- Arduino–Processing Interface: The Arduino sends angle and distance data via the serial port to the Processing IDE, which reads and interprets the data for visualization.

4. Real-Time Data Visualization

- Principle: Real-time visualization converts raw sensor data into graphical representations that can be interpreted instantly.
- Radar Display Concept:

- The green sweeping line represents the scanning direction.
- Red blips indicate detected objects at their respective distances and angles.
- Visualization uses trigonometric mapping to convert polar coordinates (angle, distance) into Cartesian coordinates for display.

5. Radar Principles

- **Scanning Mechanism:** Traditional radar systems detect objects by sending signals in all directions or within a specific sector. In this project, the servo-mounted sensor mimics this scanning action.
- **Detection and Range Limitations:** Ultrasonic waves are suitable for short-range detection, and their reflection depends on object size, material, and surface orientation.

6. Embedded System Integration

- The project combines hardware (Arduino, sensor, servo) and software (Arduino IDE, Processing IDE), demonstrating embedded system design principles, including sensor interfacing, data processing, and control mechanisms.

2.3 EXISTING TECHNOLOGIES / DESIGNS / METHODS

Radar technology has evolved over decades, primarily used in military, aviation, marine navigation, and industrial applications. Conventional radar systems utilize radio frequency (RF) waves to detect objects over long distances. These systems are highly accurate, capable of scanning large areas, and can operate in adverse environmental conditions. However, they require complex circuitry, high-power transmitters, and sophisticated signal processing units, making them expensive and inaccessible for educational or small-scale projects.

In contrast, the Arduino-based radar project employs ultrasonic waves instead of radio waves. Ultrasonic sensors like the HC-SR04 are cost-effective, easy to interface with microcontrollers, and provide adequate accuracy for short-range detection (up to 40 cm). By mounting the sensor on a servo motor, the system can sweep a semicircular area and detect nearby obstacles.

The project demonstrates the principle of radar detection in a simplified manner:

- **Distance Measurement:** Using time-of-flight calculations for ultrasonic waves.
- **Scanning Mechanism:** Achieved with a servo motor rotating the sensor.
- **Real-Time Visualization:** Data is sent via serial communication to the Processing IDE, where a radar-like display represents detected objects and scanning motion.

This approach allows students and hobbyists to understand radar operation without the need for expensive or complex equipment.

2.4 GAP ANALYSIS

Despite the widespread use of conventional radar systems, there are significant limitations for small-scale or educational purposes:

1. High Cost:

Commercial radar systems require expensive RF modules, antennas, and signal processing components, making them impractical for academic projects or personal experimentation.

2. Complexity of Design:

Traditional radar involves advanced electronics, calibration, and signal interpretation, which can be overwhelming for beginners or educational labs.

3. Limited Accessibility:

Due to cost and complexity, most educational institutions or hobbyists cannot implement or experiment with full-scale radar systems.

Bridging the Gap:

This project addresses these issues by designing an affordable, low-complexity radar system using:

- Arduino Uno as a microcontroller for control and data processing.
- HC-SR04 ultrasonic sensor for short-range detection.
- Servo motor for scanning motion.
- Processing IDE for real-time visualization.

By doing so, it provides a hands-on educational tool that allows users to learn radar principles, perform obstacle detection experiments, and gain experience in embedded systems integration without the cost and complexity of traditional radar systems.

Chapter 3

CONCEPTUAL DESIGN

3.1 COMPONENTS

- Arduino Uno Board

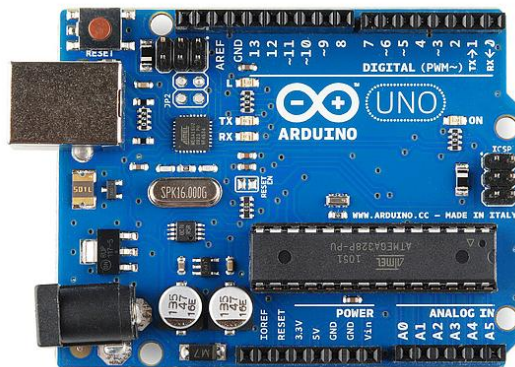


Figure 3.1.1: Arduino Uno Board

- HC-SR04 Ultrasonic Sensor



Figure 3.1.2: HC-SR04 Ultrasonic Sensor

- Servo Motor (SG90)



Figure 3.1.3: Servo Motor

- Breadboard

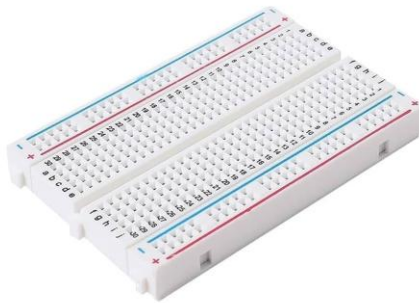


Figure 3.1.4: Breadboard

- Jumper Wires



Figure 3.1.5: Jumper Wires

- USB Cable



Figure 3.1.6: USB Cable

- Processing IDE (for visualization)

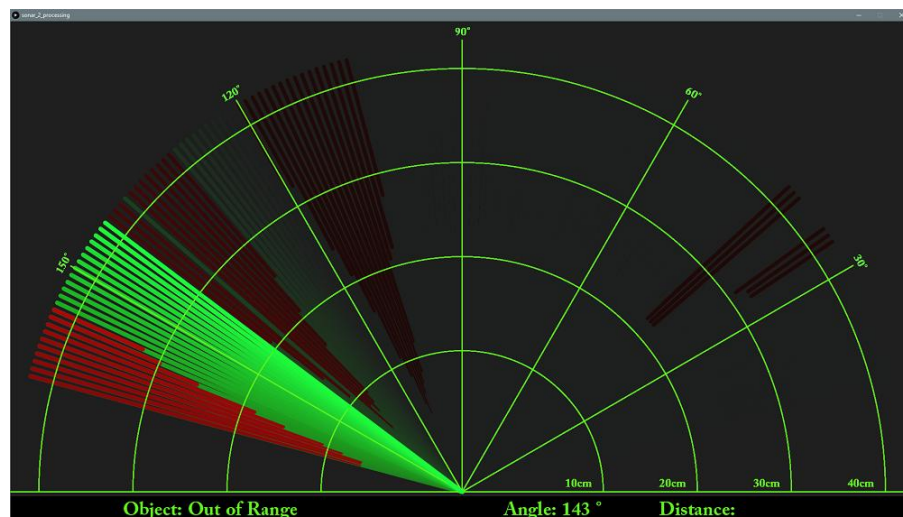


Figure 3.1.7: Processing IDE

3.2 ANALYTICAL APPROACH

The radar system operates on a combination of distance measurement and angular scanning, integrating sensor data with real-time visualization. The key analytical components are:

- Distance Calculation:

The ultrasonic sensor measures the time of flight of the emitted sound wave that reflects off an object. The distance is calculated using the formula:

$$\text{Distance (cm)} = \frac{\text{Time } (\mu\text{s}) \times 0.034}{2}$$

Here, 0.034 cm/ μ s represents the speed of sound, and division by 2 accounts for the round-trip travel of the sound wave.

- Servo Motor Sweep:

The servo motor rotates the sensor incrementally from 15° to 165°, covering a semicircular scanning area. At each angle step, the Arduino reads the distance from the sensor and transmits it to the Processing IDE for visualization.

- Data Transmission:

The Arduino communicates the angle and distance data through serial communication at a baud rate of 9600 bps. The Processing application receives this data, interprets it, and plots it on a radar-like interface in real time.

- Visualization Mapping:

The angle and distance values are converted from polar coordinates to Cartesian coordinates to display the detected objects as red blips at the correct positions on the screen.

This analytical approach ensures precise correlation between sensor data, servo angle, and visual representation, simulating the behavior of conventional radar systems in a simplified form.

3.3 CIRCUIT DIAGRAM

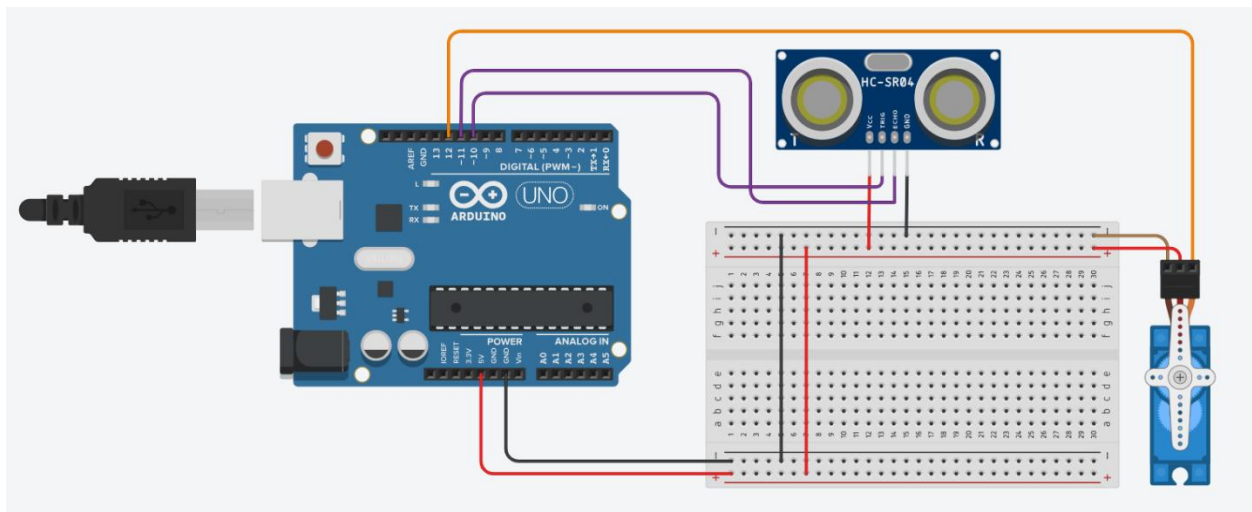


Figure 3.3: Architecture of Radar System using Tinkercad

Signal Generation:

The Arduino sends a 10 μ s HIGH pulse to the Trig pin of the ultrasonic sensor.

Echo Reception:

The sensor transmits an ultrasonic wave (40 kHz). When it hits an object, it reflects back, and the Echo pin goes HIGH for the time it takes the wave to return.

Distance Calculation:

The Arduino measures this time duration (pulseIn) and calculates distance using the formula:

$$\text{Distance (cm)} = \frac{\text{Time} \times 0.034}{2}$$

Servo Motor Rotation:

The servo motor rotates the sensor from 15° to 165° and back, scanning the area continuously.

Data Transmission:

For each angle, the Arduino sends the angle and distance values to the Processing IDE via serial communication.

Visualization:

The Processing software receives this data and plots it in real-time as a radar-like display, where:

- Green line → current scanning direction.
- Red dots → detected objects within 15–40 cm range.

3.4 SOURCE CODE

3.4.1 ARDUINO CODE

```
// Includes the Servo library
#include <Servo.h>
// Defines Trig and Echo pins of the Ultrasonic Sensor
const int trigPin = 10;
const int echoPin = 11;
// Variables for the duration and the distance
long duration;
int distance;
Servo myServo; // Creates a servo object for
controlling the servo motor
void setup() {
  pinMode(trigPin, OUTPUT); // Sets the trigPin as an
Output
  pinMode(echoPin, INPUT); // Sets the echoPin as an
Input
  Serial.begin(9600);
  myServo.attach(12); // Defines on which pin is the
servo motor attached
}
void loop() {
  // rotates the servo motor from 15 to 165 degrees
  for(int i=15;i<=165;i++){
    myServo.write(i);
    delay(30);
    distance = calculateDistance();// Calls a function for
calculating the distance measured by the Ultrasonic
sensor for each degree

    Serial.print(i); // Sends the current degree into the
Serial Port
    Serial.print(","); // Sends addition character right
next to the previous value needed later in the
Processing IDE for indexing
    Serial.print(distance); // Sends the distance value
into the Serial Port

    Serial.print("."); // Sends addition character right
next to the previous value needed later in the
Processing IDE for indexing
  }
  // Repeats the previous lines from 165 to 15
degrees
  for(int i=165;i>15;i--){
    myServo.write(i);
    delay(30);
    distance = calculateDistance();
    Serial.print(i);
    Serial.print(",");
    Serial.print(distance);
    Serial.print(".");
  }
}
// Function for calculating the distance measured by
the Ultrasonic sensor
int calculateDistance(){

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  // Sets the trigPin on HIGH state for 10 micro
seconds
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH); // Reads the
echoPin, returns the sound wave travel time in
microseconds
  distance= duration*0.034/2;
  return distance;
}
```

3.4.2 PROCESSING CODE

```
import processing.serial.*; // imports library for
serial communication
import java.awt.event.KeyEvent; // imports library
for reading the data from the serial port
import java.io.IOException;
Serial myPort; // defines Object Serial
// defubes variables
String angle="";
String distance="";
String data="";
String noObject;
float pixsDistance;
int iAngle, iDistance;
int index1=0;
int index2=0;
PFont orcFont;
void setup() {

    size (1200, 700); // ***CHANGE THIS TO YOUR
    SCREEN RESOLUTION***
    smooth();
    myPort = new Serial(this,"COM5", 9600); // starts
    the serial communication
    myPort.bufferUntil('.'); // reads the data from the
    serial port up to the character '.'. So actually it reads
    this: angle,distance.
}
void draw() {

    fill(98,245,31);
    // simulating motion blur and slow fade of the
    moving line
    noStroke();
    fill(0,4);
    rect(0, 0, width, height-height*0.065);

    fill(98,245,31); // green color
    // calls the functions for drawing the radar
    drawRadar();
    drawLine();
    drawObject();
    drawText();
}
void serialEvent (Serial myPort) { // starts reading
data from the Serial Port
    // reads the data from the Serial Port up to the
    character '.' and puts it into the String variable
    "data".
    data = myPort.readStringUntil('.');
```

```
data = data.substring(0,data.length()-1);

    index1 = data.indexOf(","); // find the character ','
    and puts it into the variable "index1"
    angle= data.substring(0, index1); // read the data
    from position "0" to position of the variable index1
    or thats the value of the angle the Arduino Board
    sent into the Serial Port
    distance= data.substring(index1+1, data.length());
    // read the data from position "index1" to the end of
    the data pr thats the value of the distance

    // converts the String variables into Integer
    iAngle = int(angle);
    iDistance = int(distance);
}
void drawRadar() {
    pushMatrix();
    translate(width/2,height-height*0.074); // moves
    the starting coordinats to new location
    noFill();
    strokeWeight(2);
    stroke(98,245,31);
    // draws the arc lines
    arc(0,0,(width-width*0.0625),(width-
    width*0.0625),PI,TWO_PI);
    arc(0,0,(width-width*0.27),(width-
    width*0.27),PI,TWO_PI);
    arc(0,0,(width-width*0.479),(width-
    width*0.479),PI,TWO_PI);
    arc(0,0,(width-width*0.687),(width-
    width*0.687),PI,TWO_PI);
    // draws the angle lines
    line(-width/2,0,width/2,0);
    line(0,0,(-width/2)*cos(radians(30)),(-
    width/2)*sin(radians(30)));
    line(0,0,(-width/2)*cos(radians(60)),(-
    width/2)*sin(radians(60)));
    line(0,0,(-width/2)*cos(radians(90)),(-
    width/2)*sin(radians(90)));
    line(0,0,(-width/2)*cos(radians(120)),(-
    width/2)*sin(radians(120)));
    line(0,0,(-width/2)*cos(radians(150)),(-
    width/2)*sin(radians(150)));
    line((-width/2)*cos(radians(30)),0,width/2,0);
    popMatrix();
}
void drawObject() {
    pushMatrix();
```

```

    translate(width/2,height-height*0.074); // moves
the starting coordinats to new location
    strokeWeight(9);
    stroke(255,10,10); // red color
    pixsDistance = iDistance*((height-
height*0.1666)*0.025); // covers the distance from
the sensor from cm to pixels
    // limiting the range to 40 cms
    if(iDistance<40){
        // draws the object according to the angle and the
distance
        line(pixsDistance*cos(radians(iAngle)),-
pixsDistance*sin(radians(iAngle)),(width-
width*0.505)*cos(radians(iAngle)),-(width-
width*0.505)*sin(radians(iAngle)));
    }
    popMatrix();
}
void drawLine() {
    pushMatrix();
    strokeWeight(9);
    stroke(30,250,60);
    translate(width/2,height-height*0.074); // moves
the starting coordinats to new location
    line(0,0,(height-height*0.12)*cos(radians(iAngle)),-
(height-height*0.12)*sin(radians(iAngle))); // draws
the line according to the angle
    popMatrix();
}
void drawText() { // draws the texts on the screen

    pushMatrix();
    if(iDistance>40) {
        noObject = "Out of Range";
    }
    else {
        noObject = "In Range";
    }
    fill(0,0,0);
    noStroke();
    rect(0, height-height*0.0648, width, height);
    fill(98,245,31);
    textSize(25);

    text("10cm",width-width*0.3854,height-
height*0.0833);
    text("20cm",width-width*0.281,height-
height*0.0833);

```

```

    text("30cm",width-width*0.177,height-
height*0.0833);
    text("40cm",width-width*0.0729,height-
height*0.0833);
    textSize(40);
    text("Indian Lifehacker ", width-width*0.875,
height-height*0.0277);
    text("Angle: " + iAngle + " °", width-width*0.48,
height-height*0.0277);
    text("Distance: ", width-width*0.26, height-
height*0.0277);
    if(iDistance<40) {
        text("      " + iDistance + " cm", width-width*0.225,
height-height*0.0277);
    }
    textSize(25);
    fill(98,245,60);
    translate((width-
width*0.4994)+width/2*cos(radians(30)),(height-
height*0.0907)-width/2*sin(radians(30)));
    rotate(-radians(-60));
    text("30°",0,0);
    resetMatrix();
    translate((width-
width*0.503)+width/2*cos(radians(60)),(height-
height*0.0888)-width/2*sin(radians(60)));
    rotate(-radians(-30));
    text("60°",0,0);
    resetMatrix();
    translate((width-
width*0.507)+width/2*cos(radians(90)),(height-
height*0.0833)-width/2*sin(radians(90)));
    rotate(radians(0));
    text("90°",0,0);
    resetMatrix();
    translate(width-
width*0.513+width/2*cos(radians(120)),(height-
height*0.07129)-width/2*sin(radians(120)));
    rotate(radians(-30));
    text("120°",0,0);
    resetMatrix();
    translate((width-
width*0.5104)+width/2*cos(radians(150)),(height-
height*0.0574)-width/2*sin(radians(150)));
    rotate(radians(-60));
    text("150°",0,0);
    popMatrix();
}

```

3.5 EXPERIMENTAL WORK

The practical implementation of the radar system involved the following steps:

1. Circuit Assembly:
 - All components were connected on a breadboard.
 - The HC-SR04 ultrasonic sensor was mounted on the SG90 servo motor.
 - The servo, sensor, and Arduino Uno were connected with appropriate wiring for power, ground, and signal pins.
2. Arduino Programming:
 - The Arduino was programmed to rotate the servo incrementally, read distance values from the ultrasonic sensor, and transmit angle-distance pairs via serial port.
3. Processing IDE Configuration:
 - The Processing IDE was installed on a laptop and configured to read serial data from COM5 port.
 - The data was parsed, and a radar-like display was generated, showing the sweeping line and detected objects.
4. Testing and Calibration:
 - The system was tested indoors to verify accuracy and response time.
 - The servo speed and delay were adjusted to ensure smooth scanning and correct visualization.

The experimental work successfully demonstrated the integration of hardware and software to create a working radar prototype.

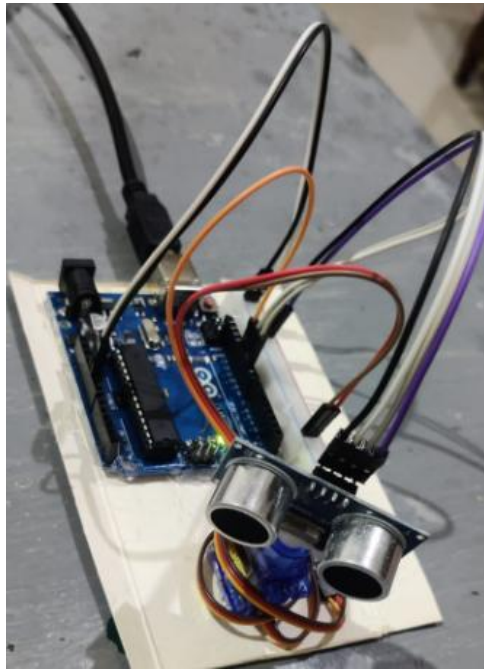


Figure 3.5.2 (a): Radar System using Arduino

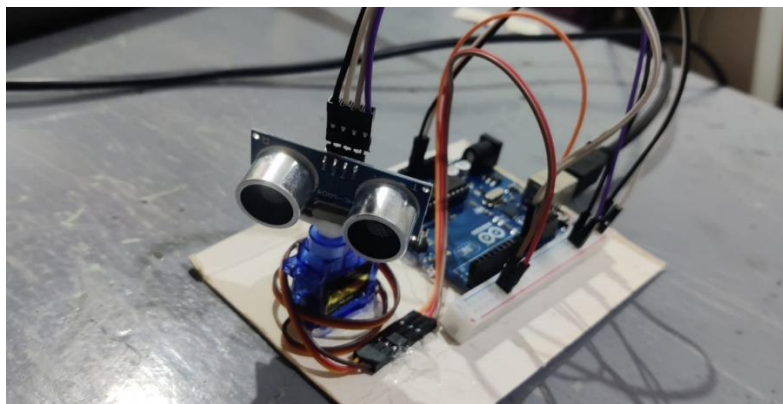


Figure 3.5.2 (b): Radar System using Arduino

3.6 CASE STUDIES

The Arduino-based radar system can be applied in multiple small-scale practical scenarios:

1. Autonomous Robots:
 - The radar can be mounted on mobile robots for obstacle detection and avoidance, enhancing navigation in unknown environments.
2. Parking Assistance Systems:
 - In miniature or model setups, the radar can assist in detecting obstacles during parking maneuvers, simulating real parking sensor functionality.
3. Security Surveillance:
 - The system can detect movement or intrusions within a defined short-range zone, providing a low-cost security monitoring solution.
4. Educational and Demonstration Purposes:
 - The project is ideal for laboratory experiments and academic demonstrations, allowing students to understand radar principles, sensor interfacing, and real-time visualization.
5. Prototype for Advanced Research:
 - This system can serve as a foundation for integrating additional sensors, wireless communication modules, or AI-based object recognition for advanced applications in IoT, robotics, and automation.

Chapter 4

RESULTS AND ANALYSIS

4.1 PRESENTATION OF DATA

The radar system collects and displays real-time data from the ultrasonic sensor as the servo motor continuously rotates within a defined angular range. The data presentation is divided into the following components:

- Servo Motor Sweep (15° – 165°):

The servo motor is programmed to rotate the ultrasonic sensor from 15° to 165° , scanning the surrounding environment in a semicircular pattern. After reaching 165° , the motor reverses its direction back to 15° , ensuring continuous scanning. This bidirectional movement allows the radar to detect objects located anywhere within that arc.

- Detection Range (Up to 15 cm Accurately):

The ultrasonic sensor (HC-SR04) measures the distance to nearby objects by emitting ultrasonic pulses and receiving their echo. The system can detect obstacles up to 15 cm accurately and can identify changes in distance as objects move closer or farther. Although the sensor can measure longer distances (up to 40 cm), the readings beyond 15 cm are less precise due to signal dispersion and reflection limitations.

- Data Transmission:

The Arduino Uno collects distance and angle data and transmits it through the serial communication port to the computer. The data includes both the angle of rotation and the distance to detected objects.

- Visual Output (Radar Display in Processing IDE):

The transmitted data is processed and displayed graphically using the Processing IDE. The visual output resembles a traditional radar interface:

- A green line sweeps across the semicircular area, representing the scanning direction.
- Red dots or blips appear on the screen at corresponding angles and distances wherever an object is detected.

- The background grid or radar circle helps visualize the range and position of objects relative to the sensor.

This presentation format makes it easy to observe, record, and analyze the radar's performance visually and in real time.

4.2 INTERPRETATION OF RESULTS

The output obtained from the radar system confirms that the designed prototype functions as intended. The results can be interpreted as follows:

- **Accurate Object Detection:** Objects located within the 15 cm range are consistently detected and displayed on the radar screen. Each object appears as a red mark, accurately positioned according to its angle and distance from the sensor.
- **Real-Time Scanning Visualization:** The moving green line on the Processing display represents the live scanning motion of the servo-mounted ultrasonic sensor. As the line sweeps back and forth, it continuously updates the radar interface, creating a dynamic and realistic visualization of the surrounding environment.
- **Consistent Communication between Hardware and Software:** The Arduino–Processing serial communication works effectively, ensuring that real-time data transfer occurs without noticeable delay. This demonstrates proper synchronization between hardware sensing and software visualization.
- **Environmental Influence on Readings:** It was observed that factors like the shape, texture, and material of objects can influence detection accuracy. Smooth, hard surfaces reflect ultrasonic waves well, producing clear signals, while soft or irregular surfaces may cause weak or distorted reflections.
- **Performance Validation:** The system performs optimally in controlled indoor environments, where noise and air disturbances are minimal. It provides a reliable demonstration of radar operation principles and validates the successful integration of all components.

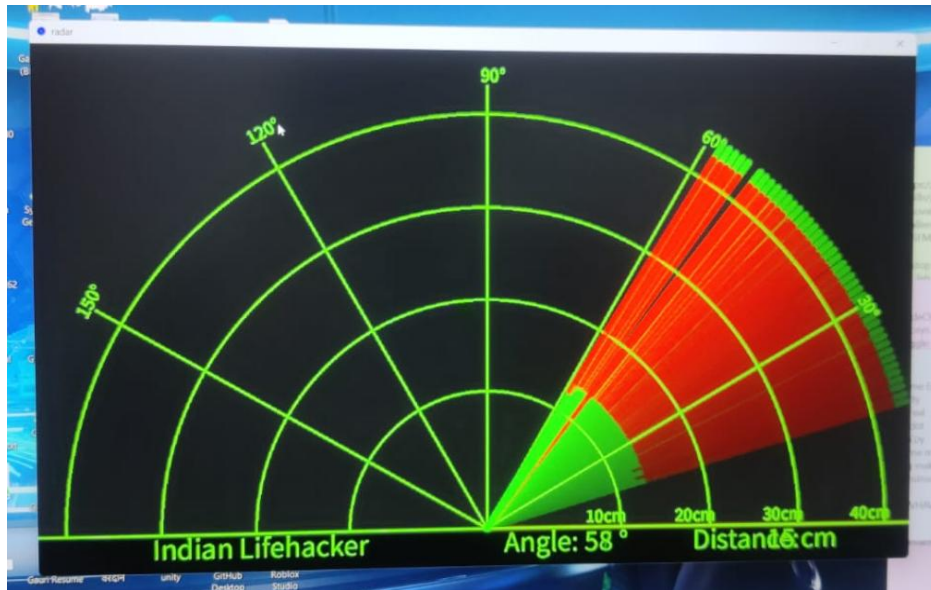


Figure 4.2.1: Visual representation of Radar System: When object is visible

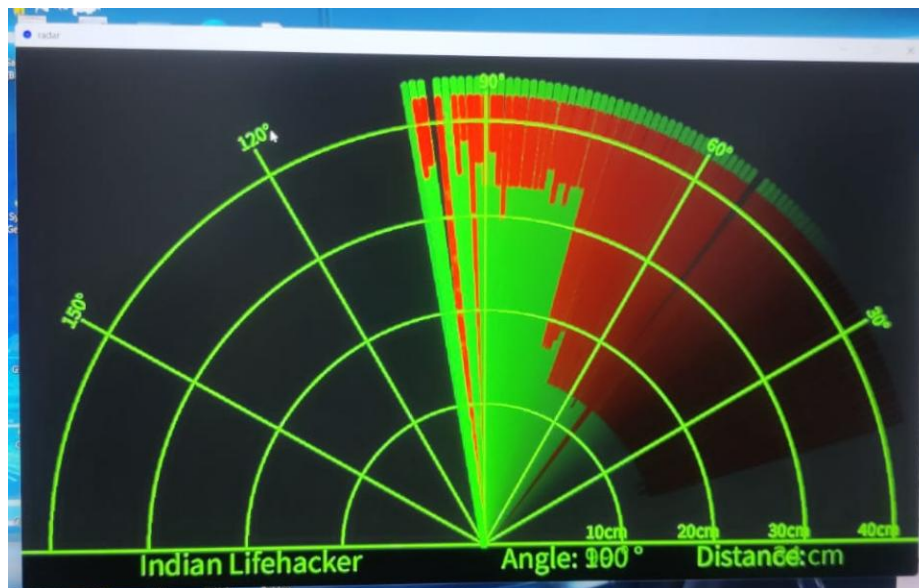


Figure 4.2.2: Visual representation of Radar System: When object is closer



Figure 4.2.3: Visual representation of Radar System: When object is away

4.3 COMPARISON

Table 4.3: Comparison between the Conventional Radar and Arduino based Radar

Feature	Conventional Radar	Arduino-based Radar
Detection Medium	Radio Waves	Ultrasonic Waves
Range	Long	Short (0–40 cm)
Cost	High	Very Low
Complexity	High	Simple
Application	Industrial/Military	Educational/Prototype

Chapter 5

DISCUSSIONS

5.1 CRITICAL EVALUATION

The developed radar system successfully demonstrates the core working principles of radar using affordable and easily available electronic components. The combination of the Arduino Uno, HC-SR04 ultrasonic sensor, and servo motor provides an effective way to detect and visualize objects in a semi-circular scanning area.

The system performs well for short-distance detection, particularly within a range of 15 cm to 40 cm. The real-time visualization using the Processing IDE makes the project interactive and easy to analyze. However, the detection accuracy depends on environmental factors such as temperature, object shape, and material type, which may affect ultrasonic wave reflection.

Overall, the project meets its objectives by offering a cost-effective, functional, and educational demonstration of radar operation. It provides valuable insight into how microcontrollers and sensors can be combined for data acquisition and visualization.

5.2 LIMITATIONS

Despite its success, the system has certain limitations that can be improved in future versions:

- **Detection Limited to ~40 cm:** The HC-SR04 ultrasonic sensor used in this project has a limited effective range. Beyond approximately 40 cm, the accuracy of detection decreases significantly, making it unsuitable for long-distance measurements.
- **Affected by Ambient Noise and Surface Material:** Ultrasonic sensors are sensitive to environmental conditions. Factors such as temperature variations, air movement, and the surface texture or angle of the target object can affect the reflection of sound waves, leading to inaccurate readings.
- **Slow Scanning due to Servo Delay:** The servo motor requires a small delay during each step of rotation to ensure stable measurements. This introduces a slight lag in scanning speed, resulting in slower data refresh rates in the radar visualization.

- **Limited Field of View:** The system scans only within 15° to 165°, which covers a semicircular area. It does not perform full 360° scanning unless additional mechanical or motor modifications are made.

5.3 IMPLICATIONS FOR FUTURE RESEARCH

This radar system serves as a basic prototype that can be enhanced in several ways for research and advanced applications:

- **Integration with Bluetooth/Wi-Fi for Wireless Monitoring:** By adding communication modules like HC-05 Bluetooth or ESP8266 Wi-Fi, the radar data can be transmitted wirelessly to smartphones or cloud platforms, enabling remote monitoring and IoT-based applications.
- **Replacing Ultrasonic with LiDAR for Better Accuracy:** LiDAR (Light Detection and Ranging) sensors provide higher accuracy, longer range, and faster response time compared to ultrasonic sensors. Integrating LiDAR would make the radar suitable for autonomous vehicles and robotic navigation systems.
- **Implementing Object Tracking and Classification using AI:** Advanced versions of this project could employ machine learning algorithms to not only detect objects but also identify and track their movement patterns. This can be applied in security surveillance, traffic monitoring, and autonomous systems.
- **360° Scanning and Multi-Sensor Integration:** Future designs can include additional servo motors or stepper motors for full-range scanning and combine multiple sensors to enhance coverage and accuracy.
- **Data Logging and Cloud Visualization:** The system can be improved to store detected data for later analysis and visualize it on web dashboards for real-time monitoring over the internet.

Chapter 6

CONCLUSION

6.1 SUMMARY OF FINDINGS

The project successfully achieved its main objective — designing and implementing a low-cost radar system using Arduino Uno, an ultrasonic sensor (HC-SR04), and a servo motor. The system effectively scans its surroundings within a specific angular range and detects obstacles located within approximately 15–40 cm.

The Processing IDE was used for real-time visualization, displaying detected objects graphically on a radar-like interface. This visual representation allows users to observe object position and distance simultaneously, closely resembling the operation of an actual radar display.

Throughout the implementation process, the system demonstrated the successful integration of hardware and software, including microcontroller programming, sensor data acquisition, servo motor control, and serial communication between Arduino and Processing.

The findings confirm that simple and affordable components can be used to replicate radar principles at a miniaturized, educational scale. Although the system has range and accuracy limitations, it effectively fulfills its purpose as a prototype for proximity detection and visualization.

6.2 CONTRIBUTIONS

This project makes several valuable contributions in both academic and practical domains:

- **Educational Value:** The project serves as a practical learning tool for students and beginners to understand how radar systems operate using embedded technology. It enhances skills in Arduino programming, sensor interfacing, and data visualization using Processing IDE.
- **Demonstration of Hardware–Software Integration:** The system illustrates how electronic components and programming can work together to achieve real-time functionality. This demonstrates the interdisciplinary nature of modern embedded systems involving electronics, programming, and graphical output.
- **Foundation for Advanced Projects:** The radar provides a base framework for developing more sophisticated systems. Future enhancements could involve wireless data

transmission, AI-based object recognition, or longer-range sensing technologies like LiDAR.

- **Practical Relevance:** The model's concept can be extended to robotics, automation, and security systems for obstacle avoidance and environmental mapping. It provides a low-cost method to test radar-like functions before implementing them in more complex systems.
- **Promotes Open-Source and DIY Learning:** By using open-source software and easily available components, the project encourages innovation, experimentation, and creativity among students and hobbyists.

Chapter 7

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